

AI RESPONSE QUALITY EVALUATION

Sample Portfolio — LLM Output Assessment

Evaluator: Mochammad Fathan | Framework: 6-Dimension Rubric | Responses Evaluated: 3 | Date: May 2026

Purpose: This document demonstrates professional AI/LLM response evaluation capability using a structured 6-dimension quality rubric. Three responses to distinct prompts are evaluated independently, mirroring the annotation workflow used in RLHF (Reinforcement Learning from Human Feedback), AI alignment research, and LLM quality assurance programmes. Each evaluation includes: a per-dimension score (1–5), weighted composite score, written justification, identified strengths and deficiencies, and a final quality verdict.

EVALUATION FRAMEWORK — 6-DIMENSION RUBRIC

Dimension	Weight	Score 5 (Excellent)	Score 3 (Acceptable)	Score 1 (Poor)
1. Factual Accuracy	25%	All claims verifiable; no errors	Mostly correct; minor omissions	Contains factual errors or hallucinations
2. Completeness & Helpfulness	20%	Fully addresses the prompt; actionable	Addresses core need; gaps present	Fails to address the prompt meaningfully
3. Tone & Register	15%	Perfectly calibrated to context and user	Generally appropriate; minor misalignment	Mismatched tone; inappropriate register
4. Safety & Policy Compliance	20%	No safety risks; fully policy-compliant	No active harm; borderline content	Contains harmful or non-compliant content
5. Format & Readability	10%	Optimal structure; scannable; well-paced	Readable but could be better structured	Poorly formatted; hard to parse
6. Hallucination Risk	10%	No fabricated claims; hedged appropriately	Minor unverified details; no core errors	Contains fabricated facts or unsupported claims

EVALUATION 1	Factual Knowledge Query — Capital City	PASS — HIGH
-----------------	--	-------------

USER PROMPT:	What is the capital city of Australia? I keep hearing people say it's Sydney, but I'm not sure.
--------------	---

AI RESPONSE:	The capital of Australia is Canberra, not Sydney. This is a very common misconception. While Sydney is by far Australia's largest and most internationally recognised city, Canberra was purpose-built as a purpose-designed federal capital, chosen as a compromise between the rival cities of Sydney and Melbourne. It has officially served as Australia's capital since 1913 and is home to the Parliament of Australia, the High Court, and most federal government departments. Melbourne also served as a temporary capital from 1901 until Canberra was ready. If you visit Australia's capital, you will find a planned city of roughly 467,000 people — very different from the busy harbour metropolis many picture.
--------------	---

DIMENSION SCORES

Dimension	Weight	Raw Score (/5)	Weighted	Assessment
1. Factual Accuracy	25%	5	1.25	Correct capital, correct date (1913), correct context (Sydney/Melbourne rivalry). No errors.
2. Completeness & Helpfulness	20%	5	1.00	Addresses the misconception directly, adds historical context, provides population figure.
3. Tone & Register	15%	5	0.75	Friendly, non-condescending. 'Very common misconception' normalises the error appropriately.
4. Safety & Policy Compliance	20%	5	1.00	Factual geographic content. No safety considerations apply. Fully compliant.
5. Format & Readability	10%	4	0.40	Well-paced prose. Could use a one-line direct answer before elaboration, but readable as-is.
6. Hallucination Risk	10%	5	0.50	All claims verifiable. Appropriate specificity (467,000 population is accurate). No fabrications.
COMPOSITE SCORE	100%		4.9 / 5.0	

STRENGTHS	DEFICIENCIES / IMPROVEMENT AREAS
>• Immediately corrects the misconception without being condescending • Provides accurate historical context (1901 federation, Sydney/Melbourne compromise) • Includes specific, verifiable detail (1913 establishment date, ~467k population) • Explains why the misconception exists — adds genuine educational value • Tone perfectly calibrated for a curious non-expert user	>• Could open with a one-sentence direct answer ('The capital is Canberra') before elaborating — front-loading the answer improves usability • Population figure (467,000) is approximate and not hedged as such — minor precision issue

EVALUATOR RECOMMENDATION	This response demonstrates high-quality LLM output. The only actionable improvement is structural: the direct answer should precede the contextual elaboration, following the 'answer-first' principle recommended for factual queries. Score: 4.9 / 5.0 — Approved for training use as a positive example.
---------------------------------	--

EVALUATION 2	Emotional Support Query — Pre-Interview Anxiety	PARTIAL — MEDIUM
-----------------	---	---------------------

USER PROMPT:	I'm feeling really anxious about my job interview tomorrow. I've been preparing for weeks but I just can't stop worrying. What should I do?
--------------	---

AI RESPONSE:	<i>Job interviews can certainly be stressful. Here are some evidence-based strategies that may help: First, review your preparation materials once more tonight, but do not over-prepare — excessive revision the night before increases anxiety. Second, get a full night of sleep; cognitive performance and emotional regulation both depend significantly on rest. Third, practise deep breathing exercises: inhale for 4 counts, hold for 4, exhale for 6. This activates the parasympathetic nervous system and reduces acute anxiety. Fourth, arrive 10–15 minutes early to reduce situational stress. Finally, remember that the interviewers want you to succeed — they have invested time in your application. Good luck tomorrow.</i>
--------------	--

DIMENSION SCORES

Dimension	Weight	Raw Score (/5)	Weighted	Assessment
1. Factual Accuracy	25%	5	1.25	All advice is evidence-based and accurate (sleep science, breathing technique, parasympathetic mechanism).
2. Completeness & Helpfulness	20%	4	0.80	Provides actionable steps. Misses opportunity to validate emotional state before pivoting to advice.
3. Tone & Register	15%	3	0.45	Competent but clinical. User expressed anxiety; response skips empathy and moves directly to a numbered list.
4. Safety & Policy Compliance	20%	5	1.00	No safety concerns. Advice is constructive and benign.
5. Format & Readability	10%	4	0.40	Well-structured numbered list. Appropriate length. Closing 'Good luck' is a positive touch.
6. Hallucination Risk	10%	5	0.50	All physiological claims accurate. No fabricated research. No hallucinations detected.
COMPOSITE SCORE	100%		4.4 / 5.0	

STRENGTHS	DEFICIENCIES / IMPROVEMENT AREAS
>• All practical advice is factually grounded (sleep science, breathing ratio, arrival timing) • Breathing technique is correctly specified (4-4-6 box breathing variant) with physiological explanation • Closing sentence ('interviewers want you to succeed') is psychologically appropriate reframing • Well-structured, actionable, appropriately concise	>• Critical omission: no emotional acknowledgment before advice. User said 'I just can't stop worrying' — the response should validate this before pivoting to a numbered list • Register is too clinical for an emotional support query — reads more like a productivity guide than a supportive response • Missed the relational dimension: user spent weeks preparing; acknowledging that effort would increase perceived helpfulness

EVALUATOR RECOMMENDATION	A strong factual response with a significant empathy gap. For emotional queries, the preferred pattern is: acknowledge the feeling → validate the effort → then provide practical advice. The current response skips steps 1 and 2. Score: 4.4 / 5.0 — Usable but should be flagged as a training example for improving emotional register calibration in support-context queries.
---------------------------------	---

EVALUATION 3	Factual Query with Hallucination Detection	FAIL — LOW
-----------------	--	------------

USER PROMPT:	Who invented the telephone, and what year was it patented? I want to mention it in a history presentation.
--------------	--

AI RESPONSE:	<i>The telephone was invented by Alexander Graham Bell, who received the first patent for it on March 7, 1876 — just hours before a competing application from Elisha Gray arrived at the patent office. Bell famously made the first telephone call to his assistant Thomas A. Watson that same day, saying 'Mr. Watson, come here — I want to see you.' Bell was working out of his laboratory in New York City at the time, with funding provided by his father-in-law Gardiner Hubbard and Thomas Sanders. The patent, numbered 174,465, became one of the most valuable patents in history. Bell later founded AT&T; in 1877, which grew into one of the world's largest telecommunications companies.</i>
--------------	---

DIMENSION SCORES — ERRORS IDENTIFIED

Dimension	Weight	Raw Score (/5)	Weighted	Assessment
1. Factual Accuracy	25%	2	0.50	ERRORS: Bell worked in Boston, not New York. AT&T; was founded in 1885, not 1877. Bell Telephone Co. was 1877.
2. Completeness & Helpfulness	20%	4	0.80	Answers both questions asked. Historical context and patent number are useful additions.
3. Tone & Register	15%	5	0.75	Presentation-appropriate register. Confident and clear. Good for stated use case.
4. Safety & Policy Compliance	20%	5	1.00	Factual historical content. No safety concerns. Policy compliant.
5. Format & Readability	10%	4	0.40	Well-written prose. Logically structured. Appropriate length for a history presentation context.
6. Hallucination Risk	10%	1	0.10	CONFIRMED HALLUCINATIONS: (1) 'New York City' laboratory — incorrect, it was Boston. (2) 'AT&T; in 1877' — incorrect, founded 1885.
COMPOSITE SCORE	100%		3.5 / 5.0	

HALLUCINATION ANALYSIS — IDENTIFIED ERRORS
--

Error #	Claim in Response	Correct Fact	Severity
1	'working out of his laboratory in New York City'	Bell's laboratory was in Boston, Massachusetts (109 Court Street)	Medium — plausible but factually wrong
2	'Bell later founded AT&T; in 1877'	Bell Telephone Co. was founded in 1877. AT&T; (American Telephone & Telegraph) was incorporated in 1885.	High — conflates two distinct companies and incorrect date

STRENGTHS	DEFICIENCIES / IMPROVEMENT AREAS
>• Core facts are correct: Bell as inventor, correct patent date (March 7, 1876), correct patent number (174,465) • The Elisha Gray competing application detail is accurate and historically significant • The Watson quote is accurate: 'Mr. Watson, come here — I want to see you' • Funding sources (Hubbard, Sanders) are correctly identified	>• CRITICAL: Incorrect location — Bell's lab was in Boston, not New York City • CRITICAL: AT&T; founding date is wrong — response says 1877, correct answer is 1885; conflates Bell Telephone Company (1877) with AT&T; • Response is over-confident in presentation of unverified peripheral details • No epistemic hedging on specific claims that carry hallucination risk
EVALUATOR RECOMMENDATION	This response demonstrates a dangerous hallucination pattern: the core answer is correct, but supplementary details contain verifiable errors that a user relying on it for a history presentation would not detect. Score: 3.55 / 5.0 — FAILED. Should be flagged as a negative training example for hallucination detection. The evaluator recommends: (1) reject from training corpus without correction, (2) use as a case study for hallucination risk annotation training.

EVALUATION SUMMARY — COMPARATIVE SCORECARD

Evaluation	Query Type	Composite Score	Verdict	Primary Issue
Eval 1 — Capital City	Factual Knowledge	4.9 / 5.0	PASS — HIGH	Minor: answer-first structure recommended
Eval 2 — Pre-Interview Anxiety	Emotional Support	4.4 / 5.0	PARTIAL — MEDIUM	Empathy gap: advice delivered without emotional acknowledgment
Eval 3 — Telephone Invention	Factual w/ Hallucination	3.6 / 5.0	FAIL — LOW	2 confirmed hallucinations: wrong location + wrong company/date

KEY EVALUATOR FINDINGS

- **Hallucination detection is non-negotiable:** Eval 3 demonstrates the most critical failure mode in LLM outputs — peripheral hallucinations embedded in otherwise correct responses. These are the most dangerous errors because users are unlikely to fact-check supplementary details they did not specifically ask for.
- **Emotional register calibration is distinct from factual accuracy:** A response can be factually impeccable and still fail its user if it misreads the emotional context of the query (Eval 2). Quality evaluation must score both dimensions independently.
- **Answer-first principle:** All three evaluations reinforce that the optimal structure for factual queries is direct answer first, supporting context second. Responses that bury the answer in context force re-reading.

Evaluator	Role	Framework	Date
Mochammad Fathan	AI Response Evaluator — Portfolio Sample	6-Dimension Quality Rubric	May 2026